

Module 2

Comparison of two means

- The t test to compare two means
- The F test to compare two variances

Comparison between the means of two **variables**

The μ and σ are population characteristics. Therefore, to know their values, you will need to compute them based on all the data points from the population.

- Not feasible to do; thus, **we will always consider** that μ and σ are unknown.

We make conclusions about μ and σ based on their sample counterparts:

$\bar{X}$: the sample mean

S : the sample standard deviation

In this module, we will study how to make conclusions about whether the **population means of two variables** (μ_1 and μ_2) are the same or not, based on the **means of two samples** ($\bar{X}_1$ and $\bar{X}_2$) taken from these two variables.

Comparison between the means of two variables

STEPS

- a) Take a sample from each variable.
- b) Obtain the sample mean for each sample (and the sample SD too).
- c) Obtain the difference between both sample means and decide whether their difference is STATISTICALLY significant for us reasonable conclude that, at the population level, the means are indeed different.

To know whether the difference between both sample means is STATISTICALLY significant, we can conduct a hypothesis test (or compute a confidence interval (NOT COVERED IN THIS COURSE)).

Hypothesis test for the
difference between two
population means: *the two
samples t-test.*

Steps of a hypothesis test

- 1) Set up H_0 and H_a
- 2) Compute the test statistic
- 3) Apply a decision rule (rejection region or P value)
- 4) Make decision and conclude

Ho and Ha

- A **hypothesis** is a statement about a characteristic (e.g., mean, std dev, etc) of one or more variables. For example, a hypothesis could be *“the mean sodium intake of someone living in the US is at most the 2,300 mg recommended by the USDA.”*
- Because the statement made about the variable (or variables) may not be supported by the data, usually a **counter statement** is made together with the **main statement**.
- The two statements are the null hypothesis (H_0) and the alternative hypothesis (H_a).

Ho and Ha

- **Null hypothesis (Ho):** The statement that represents the **status quo**. The assertion that we want to make if the **data do NOT support** the suspected **change, effect, or undesirable result**.

Ex: **Ho:** $\mu \leq 2300$

Ho represents the case where the mean sodium intake meets the USDA standards (= the status quo = the “no issue” (“no undesirable result”) situation).

- **Alternative hypothesis (Ha):** The statement that represents the assertion that we are finding evidence for (i.e., the statement referring to a change or undesirable situation).

Ex: **Ha:** $\mu > 2300$

There is a problem because the average sodium intake in US exceeds the recommended value of 2,300 mg.

Step 1: Set up H_0 and H_a

In every hypothesis test we perform, we need to choose **one** of the **three possible pairs** of H_0 and H_a (**only one pair** is applicable).

$$H_0: \mu_1 \leq \mu_2$$

$$H_a: \mu_1 > \mu_2 \quad (\text{One-tailed hypothesis (right tailed)})$$

$$H_0: \mu_1 \geq \mu_2$$

$$H_a: \mu_1 < \mu_2 \quad (\text{One-tailed hypothesis (left tailed)})$$

$$H_0: \mu_1 = \mu_2$$

$$H_a: \mu_1 \neq \mu_2 \quad (\text{Two-tailed hypothesis})$$

Step 2: Compute the test statistic

Case 1: We do NOT know whether the variances of both populations are similar or not (i.e., we assume both populations have different variances).

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- $\bar{x}_1$: Sample mean of Group 1.
- $\bar{x}_2$: Sample mean of Group 2.
- $\bar{x}_1 - \bar{x}_2$: Observed difference between the sample means.
- s_1^2 : Sample variance of Group 1.
- s_2^2 : Sample variance of Group 2.
- n_1 : Sample size of Group 1.
- n_2 : Sample size of Group 2.
- $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$: Standard error of the difference between the sample means.

The t value obtained from this formula is a value from a **t probability distribution**.

Step 2: Compute the test statistic

Case 2: We have evidence indicating that the variances of both populations are similar (i.e., we assume both populations have the same variances).

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

- $\bar{x}_1$: Sample mean of Group 1.
- $\bar{x}_2$: Sample mean of Group 2.
- $\bar{x}_1 - \bar{x}_2$: Observed difference between the sample means.
- s_1^2 : Sample variance of Group 1.
- s_2^2 : Sample variance of Group 2.
- n_1 : Sample size of Group 1.
- n_2 : Sample size of Group 2.
- s_p^2 : Pooled sample variance (combined estimate of the common population variance).
- $\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$: Standard error of the difference between the sample means.

The t value obtained from this formula is a value from a **t probability distribution**.

Step 3: Decision rule

Compare P Value (PV) to alpha

P Value formula (NO NEED to know)

$$H_a: \mu_1 - \mu_2 \neq 0$$

If $TS < 0$, then P value = 2* Area to left of the TS

If $TS > 0$, then P value = 2* Area to right of the TS

$$H_a: \mu_1 - \mu_2 > 0$$

Area to right of the TS

$$H_a: \mu_1 - \mu_2 < 0$$

Area to left of the TS

Step 3: Decision rule

Compare P Value (PV) to alpha

P Value rule

- If $PV < \alpha$ (the significance level) $\Rightarrow H_0$ is rejected and H_a supported.
- If $PV \geq \alpha$ (the significance level) $\Rightarrow H_0$ should not be rejected and H_a should not be supported.

Understanding the P Value

Reminder: In Stats, before collecting the data and running the statistical test, we assume that H_0 is true.

- Similarly to the US criminal system, where, before collecting and presenting evidence against the accused, the assumption is that s/he is NOT GUILTY.

PV definition (intuitive): The PV is a measure of how compatible the sample data is with H_0 .

- If the PV is too small, we tend to consider that the data is not compatible with H_0 (not compatible = the data refutes H_0). Small PV= Low compatibility btw data and H_0 = Evidence that H_0 might not be true.

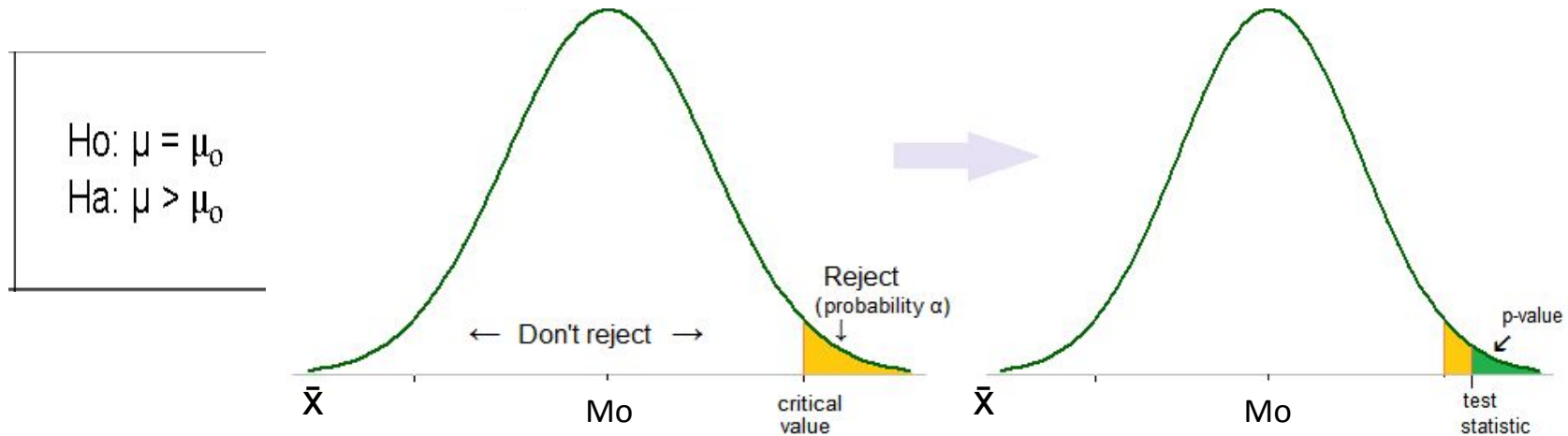
Analogy to criminal system: If the evidence is incompatible with the idea that the person is not guilty, then we declare him/her guilty.

- What do we mean by the PV being too small? Here is where the concept of alpha becomes important !

Understanding the P Value

Alpha definition (intuitive): Assuming that H_0 is true, what is the maximum risk (measured as a probability) that we want to take when we decide to reject H_0 .

- If H_0 were true, alpha is the limit that **we establish** for the PV to be considered small.



- If $P \text{ value} < \alpha$, reject H_0 and conclude that H_a is supported by data.
- If $P \text{ value} \geq \alpha$, Do not reject H_0 . Data do not support H_a .

Understanding the P Value (read independently)

PV definition (technical): If H_0 were true, what is the probability of obtaining a test statistic value that is as extreme as the one that we obtained?



Data Science Fact
@DataSciFact

...

A small p-value means the data you saw were unlikely, assuming the null hypothesis AND all your implicit assumptions.

- What PV we consider “small” depends on alpha
- Because we reject H_0 only when the PV is small, we know that the probability that we are wrong when we reject H_0 is low. That’s our protection as Data Analysts!

Unlikely != Impossible
... A small PV does mean that **H_0 is wrong** (... it means that the data does not match very well with H_0) But H_0 might be true, even if we got a small PV.

Condition for the validity of the formulas and procedure used to conduct the hypothesis test to compare the means of two variables

The probability distribution of both variables is the Normal probability distribution.

If the previous condition is not satisfied BUT the samples sizes n_1 and n_2 are at least 30, we can safely use the t-test to compare two population means.

Continue with the “**two_groups_means_nbk_1**” notebook

Statistical Significance *versus* Practical Significance

In the words of famous statisticians/ researchers:

*Statistical significance is the least interesting thing about the results. You should describe the results in terms of measures of magnitude –not just, **does a treatment affect people**, but how **much does it affect them**.* (Gene V. Glass)

*The primary product of a research inquiry is one or more measures of **effect size**, **not P values**.* (Jacob Cohen)

Reference: “Using Effect Size—or Why the *P* Value Is Not Enough” by
Gail M. Sullivan, MD, MPH and Richard Feinn, PhD

Statistical Significance *versus* Practical Significance

Statistical significance

- Does the difference that we observe in the sample can be reasonably expected to occur at the population level too? If the answer is Yes, then, the result is statistically significant
- Can the difference that we observe in the sample be **attributed to be pure randomness** or is it **because of a real difference that exists at the population level**? If the answer is the latter, then, the result is statistically significant

Practical significance

- Does the difference matter in practice? Does the difference have any practical implications?
- Given that we know the difference is real (i.e., it is statistically significant), does ***the magnitude of the difference*** have any practical consequences?

Important: Talking about practical significance only makes sense if we have already shown that there is statistical significance. Statistical significance is a necessary condition for the existence of practical significance.

Statistical Significance *versus* Practical Significance

Example: Rating of domestic guests VS rating of foreign guests

Domestic sample mean rating = 44.9 Foreign sample mean rating = 43.1

- We concluded that the difference between these sample means is statistically significant. It is a **real** difference (= if we collect data from many other samples in the future, we should keep observing a similar difference).
- But is this real difference of around 2 points important in practical terms?

We can **get an idea** (i.e., a **general sense**) of how practically significant this difference is by ***judging its magnitude***.

The magnitude of the difference can be judged as small, medium, or large. The larger the magnitude is judged to be, the more likely domain experts are to consider it impactful in practical terms (e.g., with cost, logistical, reputational, or advertising implications).

Statistical Significance *versus* Practical Significance

- We evaluate statistical significance by using the PV from a hypothesis test.
 - We evaluate the practical significance by combining (if possible) these methods:
 - a) Have people with domain knowledge evaluate the difference between the sample means and let them decide whether the difference have any practical importance.
 - b) By computing and interpreting the effect size (ES). The **ES is only a proxy** for practical significance.
- Effect size:** A parameter that tells us the extent/ degree to which the null hypothesis is wrong.

Statistical Significance *versus* Practical Significance

- The effect size calculation changes depending on the statistical test.
- For a ***test to compare the means of two populations***, the effect size formula is:

Table 1

ES Indexes and Their Values for Small, Medium, and Large Effects

Test	ES index
1. m_A vs. m_B for independent means	$d = \frac{m_A - m_B}{\sigma}$

Small (ES ~ 0.2), Medium (~ 0.5), Large (~ 0.8)

An estimation of the effect size could be:

The **absolute value** $[(\bar{X}_1 - \bar{X}_2)] / Sp$

Sp is the pooled standard deviation.

Hypothesis test for the
difference between two
population variances (or
standard deviations)

Test to compare two variances

The need to compare the average value of two variables extends to the need to compare their variability (variance, SD).

There are many real-life applications for the test to compare two variances, but the main purpose of this test in our class is to know whether we can assume the population variances to be equal ***while doing the t test.***

Test to compare two variances

Remember that we have two options while doing the **t test**:

Option 1 (when the populations *variances are assumed to be equal*)

Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Option 2 (when the populations *variances are assumed to be different*)

Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

We can apply the test to compare two variances/SDs to decide which one of these two assumptions make sense.

Test to compare two variances

Hypotheses

Two-Tailed Test

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_A : \sigma_1^2 \neq \sigma_2^2$$

For the purpose of this class, we only care about this pair of hypotheses (although there are two other pairs that could also be studied).

Test to compare two variances

Test statistic

What sample quantities should give us relevant information whether the population variances are the same?

The sample variances $(S_1)^2$ and $(S_2)^2$

Instead of computing the difference between the sample variances, we use their ratio. How should the ratio of the sample variances be to indicate the population variances are the same?

Test statistic: $F = \frac{s_1^2}{s_2^2}$ F is a variable with F distribution with n_1-1 and n_2-1 degrees of freedom.

Test to compare two variances

P VALUE (NO NEED to know)

- If $TS < 1$: Area to the left of TS + Area to the right of $1/F$ where $1/F$ follows F with n_2-1 and n_1-1 .
- If $TS > 1$: Area to the right of TS + Area to the left of $1/F$ where $1/F$ follows F with n_2-1 and n_1-1 .

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